

R SUMANTH SIMHA

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Profile Summary

Data Science undergraduate with a strong interest in Big Data Engineering, Generative AI, and Machine Learning. Passionate about solving real-world problems through distributed computing, intelligent systems, and data-driven engineering.

Education

Bellary Institute of Technology and Management, Bellary

B.E in Computer Science Engineering and Data Science

2023 - 2027

9.52/10 CGPA

BEST college, Bellary

2nd PUC

2021 - 2023

88%

Projects

Spark NYC Taxi Big Data Analytics

- Built an Apache Spark data processing pipeline to process and analyze over **213M+** NYC taxi trip records across **64 Parquet datasets**.
- Engineered scalable data-cleaning workflows, removing **26M+ invalid records** and resolving schema inconsistencies across multi-year data partitions.
- Performed large-scale trend and revenue analysis on **\$4.7B+ transaction data**, uncovering demand patterns, peak travel hours, and operational insights via a **Power BI** dashboard.
- Developed anomaly-detection logic to identify and classify **2.4M+ suspicious and corrupted transactions** using statistical and domain-based validation.
- Skills:** Apache Spark, PySpark, Big Data Analytics, ETL, Parquet, Power BI, Python

Industrial Knowledge Graph & GraphRAG Platform

- Engineered an industrial **Knowledge Graph** by integrating AI4I predictive maintenance data and technical manuals into **Neo4j**, creating over **300K+ nodes** and **510K+ relationships**.
- Built a scalable **ETL pipeline** to transform heterogeneous datasets into a unified graph schema, modeling machines, components, sensors, operating conditions, failure modes, and sensor readings.
- Developed a **GraphRAG** pipeline that combines LLM-generated Cypher queries with graph retrieval, enabling natural-language querying over industrial maintenance and operational data.
- Designed a modular architecture with automated graph ingestion, Cypher validation, retrieval, and answer generation, improving maintainability and supporting extensible industrial analytics workflows.
- Skills:** Neo4j, Knowledge Graphs, GraphRAG, Python, Ollama local LLMs

EvalGenAI

- Designed an **agentic RAG** system with a custom router that dynamically directs queries to one of **two dedicated RAG pipelines** – an Answer RAG and a Question-Paper RAG.
- Built module-aware retrieval to filter and ground context by subject module, improving response relevance for domain-specific academic queries.
- Implemented automated, structured exam question-paper generation from ingested study material, supporting multiple formats and modules.
- Architected the system using vector databases and embedding models for retrieval, integrated with LLMs for generation.
- Skills:** Python, RAG, Agentic AI, LLM Integration, LangChain, Vector Database

Technical Skills

Languages: Python, Java, SQL

Big Data & Analytics: Apache Spark, PySpark, Spark SQL, Power BI

Clouds & Databases: Supabase, MySQL, MongoDB

Frameworks: Numpy, Pandas, Tensorflow, Scikit learn, Matplotlib, LangChain

Developer Tools: Jupyter Notebook, Colab, VS Code, GitHub, Kaggle

Certifications

IBM Data Science Professional Certificate

AWS Cloud Computing Virtual Internship

Achievements

Academic Topper of the CSE–Data Science branch for the academic year **2023–2024**.

Academic Topper of the CSE–Data Science branch for the academic year **2024–2025**.

Secretary, AgentForce Club — Coordinating technical events (coding, debates) and promoting Salesforce Trailhead badge completion among members.